Ch2

一. 积(内积外积)

① 内积: $\vec{A}\cdot\vec{B} = A\cdot B\cdot\cos\theta_{AB}$ ← AB间矢量夹角(从同一次上出发)

② 外积(叉乘): $\vec{A}\times\vec{B} = \vec{a}_{AB}(A\cdot B\cdot\sin\theta_{AB}) = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

$|\vec{A}\times\vec{B}| = |A\cdot B\cdot\sin\theta_{AB}|$

方向用右手螺旋判断

相关性质:

- $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$
- $\vec{A}\times(\vec{B}\times\vec{C}) \neq (\vec{A}\times\vec{B})\times\vec{C}$
- $\vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A}) = \vec{C}\cdot(\vec{A}\times\vec{B})$
- $\vec{A}\times(\vec{B}\times\vec{C}) = \vec{B}(\vec{C}\cdot\vec{A}) - \vec{C}(\vec{A}\cdot\vec{B})$

二. 坐标

1. 矩形坐标 rectangular coordinates $(x, y, z; \vec{e}_x, \vec{e}_y, \vec{e}_z)$

① $\vec{A} = A_x\vec{a}_x + A_y\vec{a}_y + A_z\vec{a}_z$

② $\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A} = A_x B_x + A_y \cdot B_y + A_z \cdot B_z$

③ $\vec{A}\times\vec{B} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

④ $d\vec{l} = \vec{a}_x dx + \vec{a}_y dy + \vec{a}_z dz$

⑤ $d\vec{S}$: $\begin{cases} d\vec{S}_x = \vec{a}_x dy dz \\ d\vec{S}_y = \vec{a}_y dz dx \\ d\vec{S}_z = \vec{a}_z dx dy \end{cases}$

⑥ $dV = dx dy dz$

2. 柱形坐标 cylindrical coordinate $(r, \varphi, z; \vec{e}_r, \vec{e}_\varphi, \vec{e}_z)$

① $\vec{A} = A_r \vec{a}_r + A_\varphi \vec{a}_\varphi + A_z \vec{a}_z$

② $d\vec{l} = \vec{a}_r dr + \vec{a}_\varphi (r \cdot d\varphi) + \vec{a}_z dz$

③ $d\vec{S}$: $\begin{cases} d\vec{S}_r = \vec{a}_r (r \cdot d\varphi) \cdot dz \\ d\vec{S}_\varphi = \vec{a}_\varphi dr dz \\ d\vec{S}_z = \vec{a}_z (r \cdot d\varphi) dr \end{cases}$

④ $dV = dr \cdot (r d\varphi) \cdot dz$

3. 球面坐标 Spherical coordinate $(r, \theta, \varphi; \vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi)$ (θ: $(0, \pi)$; φ: $(0, 2\pi)$)

① $\vec{A} = A_R \vec{a}_R + A_\theta \vec{a}_\theta + A_\varphi \vec{a}_\varphi$

② $d\vec{l} = \vec{a}_R dR + \vec{a}_\theta (R \cdot d\theta) + \vec{a}_\varphi (R \cdot \sin\theta \cdot d\varphi)$

③ $d\vec{\zeta}$:
$$\begin{cases} d\vec{S}_R = \vec{a}_R (R\sin\theta \cdot d\varphi) \cdot (R d\theta) \\ d\vec{S}_\theta = \vec{a}_\theta (R\sin\theta \cdot d\varphi) \cdot dR \\ d\vec{S}_\varphi = \vec{a}_\varphi (R d\theta) \cdot dR \end{cases}$$

④ $dV = dR \cdot R d\theta \cdot (R\sin\theta d\varphi)$

三. 三个度和三个定理

1. 梯度 gradient 标量最大空间增长率的大小和方向的矢量

$$\nabla U = \text{grad} U = \vec{a}_n \frac{dU}{dn}$$

矩: $\nabla = \vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z}$

柱: $\nabla = \vec{a}_r \frac{\partial}{\partial r} + \vec{a}_\varphi \frac{1}{r} \frac{\partial}{\partial \varphi} + \vec{a}_z \frac{\partial}{\partial z}$

$\nabla \psi = \frac{\partial \psi}{\partial x} \vec{a}_x + \cdots \vec{a}_y + \cdots$

重点: $\frac{\partial \psi}{\partial a} = (\nabla \psi) \cdot \vec{e}_a$, $\vec{e}_a = \frac{\vec{a}}{|\vec{a}|}$

↓

函数 ψ 沿矢量 $\vec{a}$ 方向的导数

球：$\nabla = \vec{a}_R \frac{\partial}{\partial R} + \vec{a}_\theta \cdot \frac{1}{R} \cdot \frac{\partial}{\partial \theta} + \vec{a}_\varphi \cdot \frac{1}{R \cdot \sin\theta} \cdot \frac{\partial}{\partial \varphi}$

2. 散度 divergence

$$\mathrm{div}\vec{A} = \lim_{\Delta V \to 0} \frac{\oint_S \vec{A} \cdot d\vec{S}}{\Delta V} = \nabla \cdot \vec{A}$$

若为0，则该场为管形场/无散场 (continuous / solenoidal)

矩：$\nabla \cdot \vec{A} = \frac{\partial Ax}{\partial x} + \frac{\partial Ay}{\partial y} + \frac{\partial Az}{\partial z}$

柱：$\nabla \cdot \vec{A} = \frac{1}{r} \cdot \frac{\partial}{\partial r}(r \cdot Ar) + \frac{1}{r} \cdot \frac{\partial A\varphi}{\partial \varphi} + \frac{\partial Az}{\partial z}$

球：$\nabla \cdot \vec{A} = \frac{1}{R^2} \cdot \frac{\partial}{\partial R}(R^2 \cdot AR) + \frac{1}{R\sin\theta} \cdot \frac{\partial}{\partial \theta}(A\theta \cdot \sin\theta) + \frac{1}{R\sin\theta} \cdot \frac{\partial A\varphi}{\partial \varphi}$

3. 高斯定理

$$\int_V \nabla \cdot \vec{A} \, dv = \oint_S \vec{A} \cdot d\vec{S}$$

4. 旋度 circulation

$$\mathrm{rot}\vec{A} = \lim_{\Delta S \to 0} \frac{\oint_L \vec{A} d\vec{l}}{\Delta S} = \nabla \times \vec{A} = \begin{vmatrix} \vec{a}x & \vec{a}y & \vec{a}z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ Ax & Ay & Az \end{vmatrix}$$

5. 斯托克斯定理 Stokes' law

$$\int_S (\nabla \times \vec{A})\, d\vec{S} = \oint_C \vec{A} \cdot d\vec{l}$$

- 梯度：描述标量场，自身是矢量
- 散度：描述矢量场，自身是标量
  - 描述矢量场的分量沿其自身方向的变化 点乘
  - 表征场的发散特性
  - 散度为零 ==> 无源场、管形场 ==> 静磁场
- 旋度：描述矢量场，自身是矢量
  - 描述矢量场的分量沿与其垂直方向的变化 叉乘
  - 表征场的旋转特性
  - 旋度为零 ==> 无旋场、保守场 ==> 静电场

6. 特殊 formula

① $\nabla f(u,v) = \frac{\partial f}{\partial u}\nabla u + \frac{\partial f}{\partial v}\nabla v$

② $\nabla \cdot (u\vec{F}) = u \cdot \nabla \cdot \vec{F} + \vec{F} \cdot \nabla u$ u is a scalar function

③ $\nabla \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot \nabla \times \vec{F} - \vec{F} \cdot \nabla \times \vec{G}$

6. 亥姆霍兹定理 Helmholtz Theorem

① 标量场梯度的旋度为零 $\nabla\times\nabla U=0$

② 矢量场旋度的散度为零 $\nabla\cdot(\nabla\times\vec{A})=0$

③ $\vec{F}=-\nabla U+\nabla\times\vec{A}$

$\nabla\times\nabla U=0$ → gradient of scalar function ⇒ 旋度为0

$\nabla\cdot(\nabla\times\vec{A})=0$ → rotation of vector function ⇒ 散度为0

ch3 静电场

1. 电荷分布 ⇒ 密度

体：$\rho_V(\vec{r})=\lim\limits_{\Delta V\to 0}\left(\frac{\Delta q}{\Delta V}\right)$

线：$\rho_l(\vec{r})=\lim\limits_{\Delta l\to 0}\left(\frac{\Delta q}{\Delta l}\right)$

面：$\sigma_S(\vec{r})=\lim\limits_{\Delta S\to 0}\left(\frac{\Delta q}{\Delta S}\right)$

2. 电场强度：单位试验电荷受到的电场力（体现上场力）

3. 库仑定理：$\vec{F}=\frac{q_1q_2}{4\pi\varepsilon_0 R^2}\cdot\vec{a}_R$　　非真空：$\varepsilon=\varepsilon_0\varepsilon_r$

ε_r ↘ 介电常数

讨论E

① 静止点电荷在空间形成的电场

$\vec{E}(\vec{r},q_1)=\frac{q_1}{4\pi\varepsilon_0 R^2}\cdot\vec{a}_R=-\frac{q_1}{4\pi\varepsilon_0}\nabla(\frac{1}{r})$

② 若点电荷移动：

$\vec{E}(\vec{r},\vec{r}';q_1)=-\frac{q_1}{4\pi\varepsilon_0}\nabla\left(\frac{1}{|\vec{r}-\vec{r}'|}\right)$

③ 复杂情况下的电场

$$d\vec{E}=\frac{dq}{4\pi\varepsilon_0}\cdot\frac{\vec{a}_r}{R^2}\qquad dq=\rho_V dV \text{ or } \sigma_S dS \text{ or } \rho_l dl$$

4. 静电场高斯定理

发散场: $\nabla\cdot\vec{E}=\frac{\rho}{\varepsilon}$

$$\int_V(\nabla\cdot\vec{E})dV=\oint_S\vec{E}\,d\vec{S}=\frac{1}{\varepsilon}\int_V\rho dV=\frac{Q}{\varepsilon}$$

当电荷分布对称时可简便运算

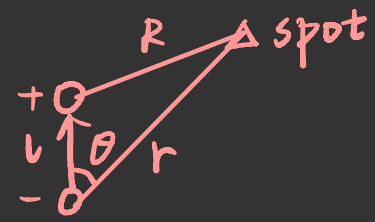
$$\frac{P_S}{2\varepsilon_0}\quad\frac{P_S}{2\varepsilon_0}$$

无旋场: $\nabla\times\vec{E}=0$

$$\oint_C\vec{E}\cdot d\vec{l}=0$$

5. 几种情况下的电场计算

① 一对同电量的正负电荷:



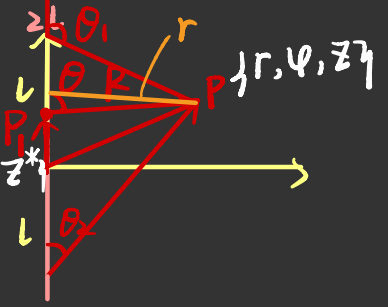
$$\vec{E}_-=-\frac{-q}{4\pi\varepsilon_0}\nabla\left(\frac{1}{|\vec{r}|}\right)$$

$$\vec{E}_+=-\frac{q}{4\pi\varepsilon_0}\nabla\left(\frac{1}{|\vec{r}|}\right)$$

$$\Rightarrow\vec{E}=\vec{E}_+ +\vec{E}_-=-\frac{1}{4\pi\varepsilon_0}\nabla\left(\frac{(q\vec{l})\cdot\vec{r}}{r^3}\right)$$

背: 和场强正比

② 一条长2L的带电导线：



法1：

$$d\vec{E}=\frac{dq}{4\pi\varepsilon_0}\cdot\frac{\vec{R}}{|R|^3}=d\vec{E}_r+d\vec{E}_z$$

$$\vec{R}=(\vec{a}_r r+\vec{a}_z z)-\vec{a}_z z^*$$

法2：

$$\vec{E}=(E_r,0,E_z),\ \rho_l=\frac{q}{2L}$$

$$d\vec{E}=\frac{\rho_l dz^*}{4\pi\varepsilon_0 R^2}(\vec{e}_r\sin\theta+\vec{e}_z\cos\theta)=\vec{e}_r dE_r+\vec{e}_z dE_z$$

$$\Rightarrow E_r=\int_{-L}^{L}\frac{\rho_l\sin\theta\,dz^*}{4\pi\varepsilon_0 R^2},\ E_R=\int_{-L}^{L}\frac{\rho_l\cos\theta\,dz^*}{4\pi\varepsilon_0 R^2}$$

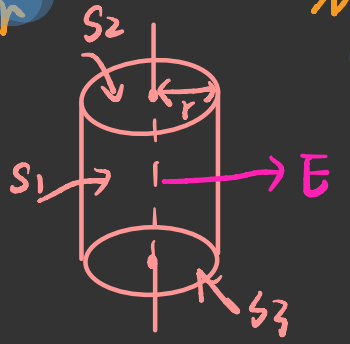
$$\because R=\frac{r}{\sin\theta},\ z^*=z-R\cos\theta\Rightarrow dz^*=\frac{r\,d\theta}{\sin^2\theta}$$

$$\Rightarrow\begin{cases}E_r=\dfrac{\rho_l}{4\pi\varepsilon_0 r}\displaystyle\int_{\theta_2}^{\theta_1}\sin\theta\,d\theta=\frac{\rho_l}{4\pi\varepsilon_0 r}(\cos\theta_2-\cos\theta_1)\\ E_z=\dfrac{\rho_l}{4\pi\varepsilon_0 r}\displaystyle\int_{\theta_2}^{\theta_1}\cos\theta\,d\theta=\frac{\rho_l}{4\pi\varepsilon_0 r}(\sin\theta_1-\sin\theta_2)\end{cases}$$

△ 若无限长：$\begin{cases}\theta_2 \to 2\pi \\ \theta_1 \to 0\end{cases} \Rightarrow \begin{cases}E_r = \dfrac{\rho_L}{2\pi\varepsilon_0}\dfrac{1}{r} \\ E_z = 0\end{cases}$

△ 若L很短（点电荷）$E = \dfrac{q}{4\pi\varepsilon_0 R^2}\cdot \vec{a}_R$

③ 一个等有无限电荷的直线



$\oint_S \vec{E}\,d\vec{s} = \dfrac{\rho_L}{\varepsilon_0}$ ×1 $= \oint_{S_1} \vec{E}\,d\vec{s} = 2\pi r E_r$ ×1

$\Rightarrow \vec{E} = \vec{a}_r E_r = \vec{a}_r \dfrac{\rho_L}{2\pi r \varepsilon_0}$

④ 导体球，电量为Q，半径为a



inside: $\vec{E} = 0$　　outside: $\dfrac{1}{\varepsilon_0}\int_V \rho\,dV = \dfrac{Q}{\varepsilon_0} = \oint_S \vec{E}\cdot d\vec{s} = E(4\pi r^2)$

$\Rightarrow \vec{E} = \dfrac{1}{4\pi\varepsilon_0 r^2} Q \vec{a}_R$

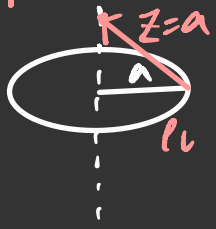
⑤ 内部全部充有电荷，体密度 ρ_0，半径为a



outside: $\vec{E} = \vec{a}_R \dfrac{Q}{4\pi\varepsilon_0 r^2} = \dfrac{\rho_0}{3\varepsilon_0}\cdot\dfrac{a^3}{r^2}$

inside: $dE = \dfrac{dq}{4\pi\varepsilon_0 r^2} = \dfrac{\rho_0 \cdot 2r' \cdot 2\pi r' \cdot dr'}{4\pi\varepsilon_0 r^2} \Rightarrow E = \int_0^r \dfrac{\rho_0 \cdot 4\pi r'^2 dr'}{4\pi\varepsilon_0 \cdot r^2} = \dfrac{\rho_0}{3\varepsilon_0} r$

⑥ 半径为a的圆形导线，线密度ρ_l，垂直于平面的轴线上 $z=a$ 处 E？

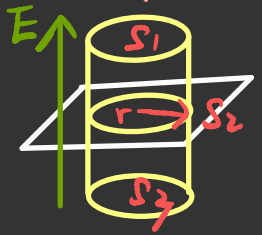


$$\vec{E}=\int d\vec{E}=\int \frac{dq}{4\pi\varepsilon_0 R^2}\cdot\vec{e}_R=\int_0^{2\pi}\frac{\rho_l d\theta\cdot a}{4\pi\varepsilon_0(a^2+a^2)}\times\frac{1}{\sqrt{2}}\times\vec{e}_z$$

$$=\frac{\rho_l}{4\sqrt{2}\pi\varepsilon_0 a}\vec{e}_z$$

注意dq可用线密度表示，注意有时只有一个方向 E

⑦ 一个无限大金属板带有ρ_s面密度电荷，板外任意一点电场强度大小？

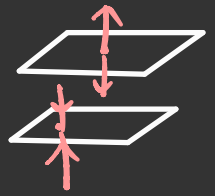


$$\oint_S \vec{E}\,d\vec{S}=\frac{\rho_S\pi r^2}{\varepsilon_0}$$

$$\oint_S \vec{E}\,d\vec{S}=\oint_{S_1}\vec{E}\,d\vec{S}_1+\oint_{S_2}\vec{E}\,d\vec{S}_2=2\pi r^2 E_r \quad\Rightarrow |\vec{E}|=\frac{\rho_S}{2\varepsilon_0}$$

在电荷极性不改变情况下，E为同只有一个（外法线）

⑧ 两个无限大金属板带有ρ_s面密度电荷（极性相反），板外任意一点电场强度大小？



板外：$\vec{E}=0$

板内：$\vec{E}=\frac{\rho_S}{2\varepsilon_0}+\frac{\rho_S}{2\varepsilon_0}=\frac{\rho_S}{\varepsilon_0}$

⑨ 在一个球体中，电荷从 $a<r<b$ 均匀分布，$\rho_v = k$，求 E.



$\oint \vec{E} d\vec{S} = \frac{Q}{\varepsilon_0}$

① $r<a$, $\vec{E}=0$

② $a<r<b$ $\quad \oint \vec{E} d\vec{S} = \frac{Q}{\varepsilon_0} = 4\pi r^2 E_r$

$\because Q = \int_a^r \int_0^{2\pi} \int_0^{\pi} r^2 \sin\theta \, d\theta \, d\varphi \, dr = 2\pi k (r^2 - a^2)$

$\therefore E = \frac{k(r^2-a^2)}{2\varepsilon_0 r^2}$

③ $r>b$ $\quad E = \frac{k(b^2-a^2)}{2\varepsilon_0 r^2}$

b. electric potential 电势 | 电位

$\vec{E} = -\nabla\psi$

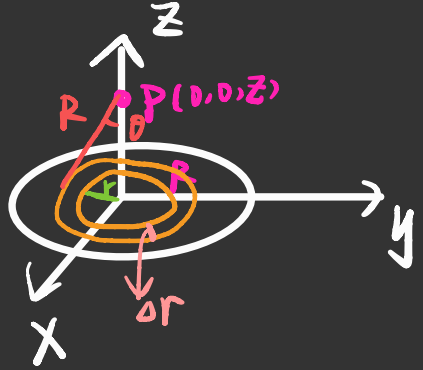
① AB间电势差：$\psi_B - \psi_A = \int_B^A \vec{E}\,d\vec{l}$

② $\psi_\infty = 0$，因此电场中点电荷的电势：$\psi_P = \frac{q}{4\pi\varepsilon_0 r}$

③ 积分化：$\psi = \int \frac{dq}{4\pi\varepsilon_0 r}$

因为 potential 是标量，计算多个点电荷形成的电场时直接代数，不用考虑方向。

例：计算圆盘上电场强度



法1. 圆盘⇒圆环

$dq = \sigma \cdot dS = \sigma \cdot [r \cdot d\varphi \cdot \Delta r]$

$dE_z = \frac{dq}{4\pi\varepsilon_0 R^2}\cos\theta$

$\cos\theta = \frac{z}{R}$

$R = \sqrt{r^2+z^2}$

$dE_z = \frac{\sigma r\,d\varphi\,dr\,z}{4\pi\varepsilon_0 R^3}$

$\Delta E_z = \int_0^{2\pi} \cdots d\varphi = \frac{r\cdot dr\cdot z\cdot\sigma}{2\varepsilon_0 R^3}$

$E_z = \int_0^a \cdots dr$

$= \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{1}{\sqrt{1+(\frac{a}{z})^2}}\right]$

法2. 采用电势

$(0,0,z)$ 的电位为 $\varphi=\int_S \frac{\rho_S ds'}{4\pi\varepsilon_0 R}$，$ds'=r'd\phi' dr'$

$$\Rightarrow \varphi=\frac{\rho_S}{4\pi\varepsilon_0}\int_0^a\int_0^{2\pi}\frac{r'dr'd\phi'}{\sqrt{z^2+r'^2}}=\frac{\rho_S}{2\varepsilon_0}(\sqrt{z^2+a^2}-|z|)$$

$$\because \vec{E}=-\nabla\varphi$$

$$=\begin{cases} e_z\frac{\rho_S}{2\varepsilon_0}\left(1-\frac{z}{\sqrt{z^2+a^2}}\right), & z>0 \\ -e_z\frac{\rho_S}{2\varepsilon_0}\left(1+\frac{z}{\sqrt{z^2+a^2}}\right), & z<0 \end{cases}$$

7. 电通量密度/电位移矢量 $\vec{D}$

$\vec{D}=\varepsilon\vec{E} \Rightarrow \nabla\cdot\vec{D}=\rho$

8. 泊松方程

$$\begin{cases}\nabla\cdot\vec{E}=\frac{\rho}{\varepsilon} \\ \vec{E}=-\nabla u\end{cases} \Rightarrow \nabla\cdot\nabla u=\nabla^2 u=-\frac{\rho}{\varepsilon}$$

△ 对于导体球

$\rho_{V_{free}}=0$ $\sigma_S\neq 0$

拉普拉斯方程

$$\nabla^2 u = 0$$

对于三个坐标系而言：

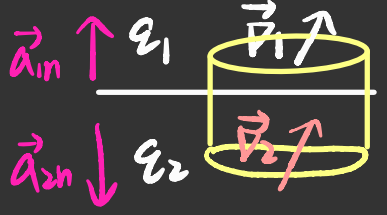
① $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

② $\nabla^2 u(r,\varphi,z) = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2}$

③ $\nabla^2 u(r,\theta,\varphi) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial u}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\cdot\frac{\partial^2 u}{\partial \varphi^2}$

9. 边界条件（boundary condition）

① 法线方向



$$\oint_S \vec{D}\cdot d\vec{S} = D_{1n}\Delta S - D_{2n}\Delta S = \rho_S \Delta S$$

$$\Rightarrow D_{1n} - D_{2n} = \rho_S$$

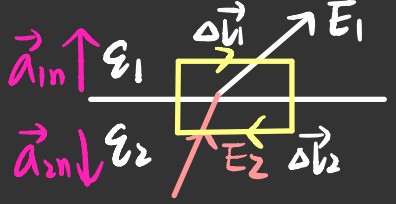
$$\vec{E} = -\nabla\Psi \Rightarrow \vec{D}\cdot\vec{a}_n = \varepsilon(-\nabla\Psi)\cdot\vec{a}_n = -\varepsilon\cdot\frac{\partial\Psi}{\partial n}$$

$$\Rightarrow \varepsilon_2\cdot\frac{\partial\Psi_2}{\partial n} - \varepsilon_1\cdot\frac{\partial\Psi_1}{\partial n} = \rho_S$$

若表面无自由电荷：$D_{1n}=D_{2n}$ ∴介质板表面不带自由电荷

若边界为导体表面：$D_{1n}=\rho_s$

② 切线方向



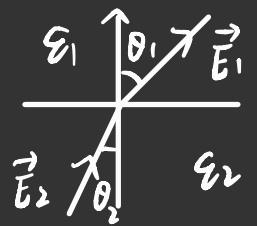
$$\oint_c \vec{E}\cdot d\vec{l} = E_{1t}\cdot\Delta l - E_{2t}\cdot\Delta l = 0$$

$\Rightarrow E_{1t}=E_{2t}$ 切向电场强度连续

$$\Delta\psi=\psi_1-\psi_2=\int_a^{a+\Delta n}\vec{E}\cdot d\vec{l}\to 0$$

$\Rightarrow \psi_1=\psi_2$ 切向电位也连续

例：介质板（ε_2）

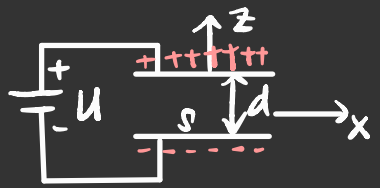


$$\frac{\tan\theta_1}{\tan\theta_2}=\frac{\varepsilon_1}{\varepsilon_2}$$

10. 电容

两个孤立导体形成的电容：$C=\frac{q}{U}$ / 一个孤立导体形成的电容：$C=\frac{q}{\varphi}$

① 例1. 平行板电容器



$$C = \frac{Q}{U} = \frac{S}{d} \cdot \varepsilon$$

②

Ch4 稳恒电流

(无散无旋)

△ 在静电场中，导体内 $E=0$，导体表面为等势面

但在稳恒电场中，… $\neq 0$ 但接近 0，…不为等势面

1. 电流密度

体：$\vec{j}=\rho\cdot\vec{v}$ (电荷 ρ × 速度)

面：$\vec{j}=\sigma_s\cdot\vec{v}$ 常用于导体

2. 恒定等式 ($\vec{j}=\sigma\vec{E}$) σ：电导率，$r=\frac{1}{\sigma}$：电阻率

① 无散

$\oint_S \vec{j}\cdot d\vec{S}=0$，$\nabla\cdot\vec{j}=0$ $(\nabla\cdot\vec{E}=0)$

② 无旋

$\oint_L \vec{E}\cdot d\vec{l}=0$，$\nabla\times\vec{E}=0$

3. 欧姆定律

① 宏观：$I=\frac{U}{R}$

② 微观：$\vec{J}=\sigma\vec{E}$

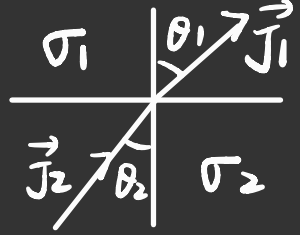
$\Rightarrow R=\frac{L}{\sigma S}$

4. Boundary conditions

切向和法向皆连续

① 法向：$J_{1n}=J_{2n}$　$\sigma_1 E_{1n}=\sigma_2 E_{2n}$　$\sigma_1\frac{\partial u_1}{\partial n}=\sigma_2\frac{\partial u_2}{\partial n}$

② 切向：$E_{1t}=E_{2t}$　$J_{1t}/\sigma_1=J_{2t}/\sigma_2$



$\frac{\tan\theta_1}{\tan\theta_2}=\frac{\sigma_1}{\sigma_2}$